

Instagram Reach Analysis

Social Media Analytics — Project Documentation Report

Project 3 • One Step Beyond the Weather Pattern Analysis

Tools: Google Sheets + Google Looker Studio • Platform: Instagram • 2026

1. Project Overview

This is the third project in a beginner data analytics learning journey. It analyses real personal Instagram data — posts and reels published between November 2025 and June 2026 — to understand reach, engagement patterns, content performance, and audience behaviour.

For the first time in this learning series, two tools are used together: Google Sheets for data collection and cleaning, and Google Looker Studio for building an interactive visual dashboard. Looker Studio is a free, browser-based tool by Google that works on any device including Mac, making it the ideal alternative to Power BI for this project.

The data was collected manually from Instagram's Professional Dashboard Insights — recording reach, impressions, likes, comments, shares, saves, and profile visits for each post individually.

2. Objectives

- Analyse the total reach and impressions generated across 17 Instagram posts and reels
- Identify which month had the highest posting activity and reach
- Find which individual post or reel achieved the best engagement rate
- Compare performance between Reels and Photo posts
- Build an interactive dashboard in Google Looker Studio with filters and scorecards
- Learn new tools and skills: Looker Studio, scorecards, drop-down filters, date range control, time series chart, and grouped bar chart

3. Dataset Description

The dataset `instagram_clean.csv` was created manually by recording Instagram Insights data for each post from the Instagram Professional Dashboard. The data was then cleaned and processed in Google Sheets before being connected to Looker Studio.

Dataset overview

Property	Detail
File name	instagram_clean.csv

Total records	17 posts and reels
Period covered	November 2025 to June 2026 (8 months)
Platform	Instagram (Professional account)
Data source	Manually entered from Instagram Professional Dashboard Insights
Content types	Reels and Photo posts

Column descriptions

Column	Example	Description
Reel_Num	1	Sequential post number from 1 to 17
Date	23-04-2026	Date the post was published (DD-MM-YYYY)
Month	April	Month name extracted from the date
Day_of_Week	Thursday	Day the post was published
Post_Type	Reel	Type of content: Reel or Photo post
Reach	4219	Number of unique accounts that saw the post
Impressions	5100	Total number of times the post was displayed (including repeats)
Likes	312	Total likes received on the post
Comments	14	Total comments received
Shares	28	Number of times the post was shared
Saves	9	Number of times the post was saved
Profile_Visits	45	Profile visits generated from this post
Engagement_Rate	16.3	(Likes+Comments+Shares+Saves) / Reach x 100. Calculated in Google Sheets.

4. Tools Used

Tool	Purpose in this project
Google Sheets	Data entry, cleaning, Engagement_Rate formula, and Total_Engagement helper column
Google Looker Studio	Interactive dashboard with scorecards, charts, drop-down filters, and date range control
Instagram Insights	Source of all raw metrics — accessed via Professional Dashboard on Instagram app
GitHub	Project repository for storing all files and sharing the Looker Studio dashboard link

This project uses two tools for the first time together — Google Sheets as the data backend and Google Looker Studio as the visualisation frontend. Looker Studio connects directly to Google Sheets, which means no CSV importing is needed and the dashboard updates automatically when the Sheet is updated.

5. New Skills Learned in This Project

This project introduces six new skills compared to the previous Weather Pattern Analysis project, with the most significant being the use of Google Looker Studio as a professional dashboard tool for the first time.

1

Google Looker Studio

A free browser-based dashboard tool by Google. Unlike Google Sheets (where charts are static), Looker Studio creates interactive dashboards where filters and controls update all visuals simultaneously. Works on Mac, Windows, and any browser without installation.

2

Scorecards (KPI Cards)

Large single-number tiles that display the most important metrics at the top of a dashboard — Total Reach, Average Engagement Rate, and Total Likes. Called scorecards in Looker Studio and cards in Power BI. Standard feature of every professional analytics dashboard.

3

Drop-down Filters

Interactive controls that let the viewer filter all dashboard charts by selecting a value from a dropdown menu. Used to filter by Month and by Post_Type (Reel or Photo). When a filter is selected, every chart on the page updates instantly.

4

Date Range Control

A special filter that lets viewers select any custom date range using a calendar picker. All charts update to show only data from the selected period. This makes the dashboard useful for any future time period as more posts are added.

5

Time Series Chart

A line chart specifically designed for date-based data in Looker Studio. The X-axis automatically handles date formatting and grouping. Used to show Reach Over Time from November 2025 to June 2026.

6

Grouped Bar Chart

A bar chart with multiple metrics shown as separate bars side by side for each dimension value. Used to compare Likes, Saves, and Shares by Post Type (Reel vs Photo) in a single chart. More informative than a simple bar chart because it shows multiple metrics together.

6. Steps Performed

Step 1 — Data Collection

Data was collected manually by opening each post and reel on Instagram, tapping the View Insights button, and recording all available metrics into the `instagram_clean.csv`

template. The template was pre-populated with dates, captions, and hashtags extracted from the Instagram JSON data download. Metrics recorded for each of the 17 posts included reach, impressions, likes, comments, shares, saves, and profile visits.

Step 2 — Data Cleaning in Google Sheets

The CSV file was imported into Google Sheets. The following cleaning steps were applied:

- Date column formatted as proper date values (not text or Excel serial numbers)
- Month and Day_of_Week columns verified for consistent capitalisation
- Post_Type column checked for trailing spaces and corrected
- Engagement_Rate column calculated using the formula:
 $(\text{Likes} + \text{Comments} + \text{Shares} + \text{Saves}) / \text{Reach} \times 100$
- Total_Engagement helper column added: $=\text{Likes} + \text{Comments} + \text{Shares} + \text{Saves}$

Step 3 — Connecting Google Sheets to Looker Studio

Looker Studio was opened at lookerstudio.google.com. A new blank report was created. The Google Sheets connector was used to link the Instagram Reach Analysis spreadsheet directly to the report, making all columns available as dimensions and metrics in the dashboard.

Step 4 — Building the Dashboard

Seven visuals were added to the dashboard canvas in the following order:

- Dashboard title text box at the top centre
- Three scorecards in a row: Total Reach (30,002), Avg Engagement Rate (16.3%), Total Likes (3,545)
- Time series line chart: Reach Over Time (Date on X-axis, Reach on Y-axis)
- Bar chart: Total Reach by Month (Month dimension, Reach metric, sorted descending)
- Column chart: Engagement Rate per Post (Reel_Num dimension, Avg Engagement_Rate metric)
- Grouped bar chart: Engagement by Post Type (Post_Type dimension, Likes + Saves + Shares metrics)
- Two drop-down filters: one for Month, one for Post_Type
- Date range control for custom date filtering

Step 5 — Formatting and Publishing

A consistent colour theme was applied using the Instagram brand colour (#E1306C pink). Charts were aligned and resized for a clean layout. The report was tested in View mode to verify all filters updated all charts correctly. A shareable public link was generated and the report was exported as a PDF.

Step 6 — GitHub Upload

All project files were organised into folders and uploaded to GitHub. The Looker Studio shareable link was added to the README.md file so anyone can view the live interactive dashboard without downloading any software.

7. Dashboard Visuals

Scorecards — KPI Summary

Three scorecard tiles at the top of the dashboard display the most important numbers at a glance: Total Reach of 30,002 across all 17 posts, Average Engagement Rate of 16.3%, and Total Likes of 3,545.

Visual 1 — Reach Over Time (Time Series Chart)

A line chart showing how reach changed from November 2025 to June 2026. A clear spike is visible in April 2026 driven by 8 posts published that month including the highest-reach reel. The chart uses a pink line colour matching the Instagram brand.

Visual 2 — Total Reach by Month (Bar Chart)

A horizontal bar chart ranking months by total reach. April dominates with 14,695 reach, followed by June and March. Months with fewer posts (November, December) show significantly lower bars, confirming that posting frequency directly impacts total monthly reach.

Visual 3 — Engagement Rate per Post (Column Chart)

A vertical column chart showing the engagement rate for each of the 17 posts individually. Post 17 (November mountains content) shows the tallest bar at 41%, significantly above all other posts. This chart reveals that reach size and engagement rate are not always correlated.

Visual 4 — Engagement by Post Type (Grouped Bar Chart)

A grouped bar chart comparing Likes, Saves, and Shares side by side for Reels versus Photo posts. This reveals whether one content type generates more passive engagement (likes) or active engagement (saves and shares).

Filters

Two drop-down filters (Month and Post_Type) and one date range control are placed at the top right of the dashboard. Selecting any value instantly filters all four charts simultaneously, making the dashboard fully interactive for any combination of month and content type.

8. Key Findings

30,002

Total Reach across 17 posts

Average of 1,765 reach per post over 8 months

16.3%

Average Engagement Rate

Industry benchmark is 3–6%. This account performs at nearly 5x the average.

4,219

Highest single post reach — April 23, 2026

Mountains reel with hashtags #mountains #nature #himalayas

41%

Highest engagement rate — November 2025

Post with only 605 reach but highly loyal and engaged audience

14,695

April 2026 total reach — best month

8 posts published in April — highest posting frequency of any month

Additional findings:

- Likes account for the large majority of total engagement at 3,545, compared to 290 shares and 117 comments, indicating the audience prefers passive engagement over active sharing
- Saves (64 total) are low relative to likes, suggesting content is appreciated in the moment but not widely bookmarked for future reference
- Posts published in November and December had the lowest reach, likely due to lower posting frequency and potentially lower audience activity in those months
- The relationship between reach and engagement rate is inverse in some cases: the highest-reach post does not have the highest engagement rate, and the highest engagement rate post has one of the lowest reach values

9. Challenges Faced

- Instagram's data download (JSON format) does not include reach, impressions, or engagement metrics. These are only visible inside the app under Professional Dashboard Insights, requiring manual data entry for all 17 posts.
- The Date column in the CSV was initially stored as Excel serial numbers after the JSON-to-CSV conversion. These had to be reformatted as proper dates in Google Sheets before Looker Studio could recognise them as a date dimension.
- Power BI Desktop was the originally planned tool but is not available on Mac. Google Looker Studio was selected as an alternative and proved to be a better fit since it connects directly to Google Sheets without any file importing.
- The Engagement by Post Type grouped bar chart required adding multiple metrics to a single chart in Looker Studio, which involved a slightly different setup than single-metric charts.

10. GitHub Repository Structure

File Path	Description
dataset/instagram_clean.csv	Clean dataset — 17 posts with all metrics
analysis/instagram_analysis.xlsx	Google Sheets export with cleaned data and formulas
dashboard/dashboard.pdf	Looker Studio dashboard exported as PDF
screenshots/dashboard_screenshot.png	Full screenshot of the interactive dashboard
docs/documentation.pdf	This project documentation report
README.md	Project overview with live Looker Studio dashboard link

Note: Unlike Power BI projects which require a .pbix file upload, Google Looker Studio dashboards are shared via a public URL. The README.md contains the live dashboard link so anyone can view and interact with the dashboard directly in their browser without downloading any software.

11. Project Progression — What I Learned

Skill Area	Project 1 (Sales)	Project 2 (Weather)	Project 3 (Instagram)
Tool	Google Sheets	Google Sheets	Sheets + Looker Studio

Data source	Kaggle CSV	Historical CSV	Own Instagram data
Key formula	SUMIF	AVERAGEIFS	Engagement Rate calc
Visualisation	Sheets charts	Combo chart, heatmap	Interactive dashboard
New feature	Pivot Tables	Sparklines, freeze	Filters, scorecards
Dashboard type	Static charts	Static with heatmap	Fully interactive

12. Conclusion

This project successfully analysed 17 Instagram posts and reels using Google Sheets and Google Looker Studio, producing a fully interactive dashboard with scorecards, time series charts, bar charts, and real-time filters.

The key finding is that content quality matters more than posting frequency for engagement. The November 2025 post achieved a 41% engagement rate despite having only 605 reach, while April 2026 dominated in total reach due to high volume posting. Mountains and nature content consistently performed better than other themes, which suggests a content strategy direction for future posts.

Using Looker Studio for the first time introduced six new professional skills: scorecards, drop-down filters, date range controls, time series charts, grouped bar charts, and dashboard sharing via public URL. These skills are directly transferable to real-world business analytics roles where Looker Studio and similar tools (Power BI, Tableau) are widely used.

This project also demonstrated the complete data analytics workflow for the first time using personal real-world data: data collection, cleaning, analysis, visualisation, and presentation — making it the most end-to-end project in this learning series so far.

End of Documentation

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